

Nathaniel Murcin

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Aerospace Engineering — GPA: 4.00 (Faculty Honors)

Aug. 2023 – May 2027

- Relevant Coursework: Rocket Propulsion, Thermodynamics, Structural Analysis, System Dynamics and Controls

EXPERIENCE

Pursuit Aerospace

Newburyport, MA

Engineering Intern

Aug. 2025 – Dec. 2025

- Designed, 3D modeled, manufactured, and implemented 15 quick-change dotpeen marking fixtures, reducing load/unload time from an average of 6s to 1s per part.
- Collaborated in Kaizen workshop on lock lug product line, transitioning from batch to one-piece flow and increasing machine count; designed quick-check gauges reducing inspection from 40s to 2s & cut production time by 55%.
- Developed and documented in-house supercleaning process for Rolls Royce component, including fixture design and full procedure/spec creation, reducing lead time by 3 days and saving \$1000 per part (approved for long-term use).
- Applied GD&T and process control standards across fixture and component designs, maintaining tight tolerances and ensuring repeatability through multiple test and qualification cycles.

Georgia Tech Experimental Rocketry (GTXR)

Atlanta, GA

Fin Design and Manufacturing Lead

Aug. 2023 – Present

Design & Manufacturing

- Redesigned composite fin structures and fixturing hardware following a roll-coupling failure, integrating structural and aerodynamic analysis to improve flight reliability.
- Designed, 3D-modeled and CAM programmed a precision fin alignment jig with $\pm 0.1^\circ$ cant tolerance; incorporated DFM and GD&T principles to ensure manufacturability, alignment repeatability and accuracy.
- Machined flight and ground hardware using CNC mill and waterjet and manual lathe; verified part accuracy through iterative inspection and adjustment of tool paths.

Testing & Operations

- Aided in launch procedure and integration of 2 rockets (one supersonic, one hypersonic) at FAR test site in Mojave, CA, requiring on-site rapid design, fabrication, and testing of mission critical components.
- Executed post-launch failure analysis, leading to incorporation of fin cant for stability improvement.
- Developed and executed mechanical test setups for determining in-plane shear modulus of composite structures, comparing results with analytical FEA models.

Publications

- Co-authored *Shear Modulus Testing in Composite Sandwich Panels*, published in the AIAA Journal and presented at the AIAA Region II Student Conference ([link](#)).
- Co-authored *Viability of Thermoformed Closed-Cell Foam Cores for Carbon Fiber Sandwich Rocket Fins*, presented at the AIAA Region II Student Conference and soon-to-be published in the AIAA Journal

Ben T. Zinn Combustion Laboratory

Atlanta, GA

Undergraduate Researcher

Jan. 2026 – Present

- Designed and built tubing layouts for multi-gas transport, while reducing PIV datasets for attached flame analysis.

Vertical Lift Research Center of Excellence (VLRCE)

Atlanta, GA

Undergraduate Research Assistant

Jan. 2025 – May 2025

TECHNICAL SKILLS

Software: SolidWorks, CAMWorks, ANSYS, MATLAB, Python, RASaero, LaTeX

Analysis: FEA, thermodynamics, fluid mechanics, flutter and structural analysis

Hardware & Manufacturing: CNC machining, lathe, CAM programming, GD&T, DFM, waterjet cutting, composite layups, 3D printing, mechanical testing (Instron), wind tunnel proficiency

Interests: Backpacking, Cello, Disc Golf, Strength Training, Travel, Astronomy